Rapid Revision Summary

Last-minute review · condensed from 16 papers and 17 mark schemes, Jan 2019 – Jun 2026

THE PAPER

2 hours · 110 marks · 9–11 questions · ≈ 1 min per mark. MCQs, "state the equation", calculations, a method, a graph and 3–6 mark explanations. Recall equations are **not** given.

TOP 12 MARK-LOSERS

- No unit conversion → **−1 POT error**.
- Wrong s.f. when asked (8.55 instead of 8.5).
- "Gravity" instead of **weight**; "C" for current.
- Half-life as "half the time".
- CMBR as background radiation.
- "Dangerous/harmful" with no reason.
- One-sided comparisons.
- Graph scale in 3s; missing units on axes.
- Voltmeter not in parallel with the component only.
- Anomaly included in a mean.
- Two ticks in one row.
- Bald answers: "the weather", "bigger magnet".

EQUATIONS TO RECALL

Quantity	Equation
speed	distance ÷ time
acceleration	(v − u) ÷ t
force	F = ma
weight	W = mg
work	W = Fd
power	P = W ÷ t
KE	$\frac{1}{2}mv^2$
GPE	mgh
efficiency	useful ÷ total
density	ρ = m ÷ V
pressure	p = F ÷ A
pressure difference	p = hρg
voltage	V = IR
charge	Q = It
power	P = IV
energy	E = IVt
wave speed	v = fλ
frequency	f = 1/T

Given when needed: $v^2 = u^2 + 2as$ · $v = 2\pi r/T$ · $p_1/T_1 = p_2/T_2$ (K = °C + 273). g = 10 N/kg.

FORCES AND MOTION

- v–t: gradient = a, area = distance. −45/39 = −1.2 m/s^2; ½ × 45 × 39 = 880 m.
- F = ma: 7600/1200 = 6.3; s = 21^2 ÷ (2 × 6.3) = 35 m.
- Stopping = thinking + braking (13.5 + 23.25 = 37 m).
- Thinking: tiredness, alcohol, drugs, phone, age. Braking: speed, mass, brakes, tyres, road, slope.
- Hooke: extension ∝ force to limit of proportionality. Elastic: returns to shape.
- Vector: magnitude + direction. 26 m/s × 3.6 = 94 km/h.

TERMINAL VELOCITY (4–6)

Weight > drag → accelerates → speed ↑ → drag ↑ → resultant ↓ → a ↓ → drag = weight → resultant 0 → terminal velocity.

ELECTRICITY

- 2200 ÷ 230 = 9.6 A → 13 A fuse.
- Series: voltages add (6.0 − 3.9 = 2.1 V); I = 2.1/290 = 0.0072 A; R = 3.9/0.0072 = 540 Ω.
- E = IVt: hours → seconds first.
- I–V: ammeter series, voltmeter parallel, variable resistor. Lamp: curve flattens. Diode: one way only.
- Thermistor: T ↑ → R ↓ → I ↑ → V across fixed resistor ↑.
- Parallel lights: full voltage each, brighter, independent switching.

EXPLANATIONS

Heating: resistance; electrons collide with lattice ions → transfer energy.

Fuse: I > rating → fuse melts → circuit broken → appliance isolated.

Earth: live touches case → large current to earth (low R) → fuse blows/breaker trips.

Circuit breaker: faster, resettable.

WAVES

- v = fλ: 3.0 × 10^8 ÷ 5.2 × 10^9 = 0.058 m; ÷ 170 MHz = 1.8 m.
- Echo: double the distance (2 × 4.70 ÷ 345 = 0.0136 s).
- Transverse: ⊥ to energy transfer (EM). Longitudinal: ∥ (sound).
- Sound vs light: longitudinal/transverse, needs medium/vacuum, slow/fast, speeds up/slows in denser medium.
- All EM: same speed in vacuum, transverse, transfer energy.
- Reflection: i = r from the normal (90°).

EM	Danger
microwave	internal heating
infrared	skin burns
UV	skin cancer, eyes
X-ray, gamma	ionising → mutation, cancer

ENERGY AND HEAT

- ½ × 85 × 4.5^2 = 860 J → h = 860/850 = 1.0 m.
- W = Fd: 41 000/15 = 2700 N.
- Sankey: widths to scale; waste = thermal to surroundings.
- Conduction: ion vibrations + free electrons.
- Black: best absorber/emitter of infrared.

CONVECTION (4)
Water near heater heated → expands → less dense → rises → cooler denser water sinks → cycle repeats.

SOLIDS, LIQUIDS, GASES

- Same material ⇒ same density.
- p = hρg: 15.8 × 1030 × 10 = 162 740 Pa + 100 kPa = 260 kPa.
- p/T in kelvin: 9.3 × 755/726 = 9.7 MPa.
- v = √(2KE/m) = 410 m/s.
- Hotter gas → faster particles → more frequent, harder collisions → higher pressure.

MAGNETISM

PLOT A FIELD
Compass near magnet → mark needle direction → move and repeat → join, arrows N to S. Or iron filings, tap the card.

- Motor: current → field → interacts with magnet's field → force; opposite sides opposite forces; commutator reverses current each half turn.
- Induction: coil cuts field lines → voltage. More: faster, stronger magnet, more turns.
- Steel hard (keeps magnetism); iron soft. Uniform field: parallel, evenly spaced.

RADIOACTIVITY

	Nature	Stopped by
α	He nucleus	paper
β^-	electron	few mm Al
γ	EM wave	thick lead

- β^-: n → p, A same, Z + 1. α: A − 4, Z − 2.
- Half-life: time for half the nuclei to decay / activity to halve. 3 × 5.6 = 16.8 years.
- Background: rocks, radon, cosmic, food. Mean → subtract.
- Safety: tongs, lead box, short time, large distance.

FISSION CHAIN (4)
Neutron absorbed by U-235 → splits → daughter nuclei + 2–3 neutrons + energy → neutrons absorbed by other nuclei. Graphite slows; boron absorbs; concrete shields.
Fusion: light nuclei join; needs high T and p because nuclei repel.

ASTROPHYSICS

- Sun-like: nebula → protostar → main sequence → red giant → white dwarf → black dwarf.
- Massive: red supergiant → supernova → neutron star/black hole (depends on mass).
- Colour: red coolest → blue hottest.
- v = 2πr/T: 2π × 8.7 × 10^{10} ÷ 2.5 × 10^8 = 2200 m/s (time in seconds).
- Planet: near-circular, constant speed. Comet: very elliptical, fastest near star (at a focus).
- Universe > galaxy > solar system.

PRACTICAL AND GRAPHS

- Independent, dependent, control variables; repeat, mean.
- Scales > ½ grid, no 3s; labels + units; ±½ square; best-fit curve; circle anomaly.
- Continuous data → line graph; categoric → bar chart.